

Program Schemes for BE (CSE) (Hons.) (with Specialization in Cloud Computing)

Cluster: Engineering

Institute: UIE

Department: AIT-CSE

Type of Program: UG

First Year				Second Year			
Semester-1	Course Name	Course Category	Credits	Semester-3	Course Name	Course Category	Credits
	Logical Thinking & Problem Solving	Major Core	4		Computer Organization and Architecture	Major Core	3
	Mathematics-I	Major Core	5		Advanced Data Structures	Major Core	4
	Communication Skills-I	Skill Enhancement Course	2		Agile Development Methodologies	Minor Core	4
	Design Thinking and Innovation	Skill Enhancement Course	1		Database Management System	Major Core	4
	Semiconductor Physics and Applications	Major Core	3		Object Oriented Programming using JAVA	Major Elective	2
	Universal Human values, Ethics and life skills-1	Value Education	1		Probability and Statistics	Major Core	4
	AI Applications Lab	Ability Enhancement Course	1		Universal Human Values, Ethics and Life Skills-2	Value Education	2
	Computer Graphics Lab	Major Core	1		Social Internship	Internship	2
	Biology for Engineers	Major Core	3		Aptitude-I	Ability Enhancement Course	1

	Academic & Career pathway planning	Value Education	0		Gender Equity and Empowerment	Value Education	1
					Entrepreneurship Mindset	Ability Enhancement Course	1
					Critical Enquiry and Research Methodology	Ability Enhancement Course	1
					General Proficiency-II	Ability Enhancement Course	1
Semester -2	Data Structures & Algorithms	Major Core	4	Semester-4	Operating System	Major Core	5
	Mathematics-II	Major Core	5		Computer Networks	Major Core	4
	Communications Skills - II	Skill Enhancement Course	1		Advanced Database Management System	Major Core	1
	Programming Practice	Major Core	1		Professional Elective -I	Minor Elective	4
	Digital Electronics	Major Core	4		Statistical Methods for AI	Major Core	4
	Emerging & Disruptive Technologies Workshop	Major Core	2		RIE-1	Ability Enhancement Course	1
	Engineering Physics	Major Core	4		RIE-2	Skill Enhancement Course	1
	General Proficiency-I	Ability Enhancement Course	1		General Proficiency-III	Skill Enhancement Course	1
	Soft Skills-I	Skill Enhancement Course	1		Competitive Coding-I	Skill Enhancement Course	1
					Mid-Grade Project-1	Project	1
					RIE-1		

					Employability-TPP: Aptitude-II	Ability Enhancement Course	1
					Innovation Marketing & Management Essentials	Ability Enhancement Course	1
					Research-Academic Writing	Ability Enhancement Course	1
					RIE-2		
					Employability-TPP: Soft Skills-II	Skill Enhancement Course	1
					Innovation - AI Automation & Sustainability	Skill Enhancement Course	1
					Research - Research Ethics & Intellectual Property Rights	Skill Enhancement Course	1

	Third Year				Fourth Year		
Semester -5	Course Name	Course Category	Credits	Semester-7	Course Name	Course Category	Credits
	Deployment of Private Cloud	Major Elective	4		Blockchain	Minor Core	4
	Full Stack Development-I	Minor Core	2		Professional Elective-VI	Minor Elective	3
	Design and Analysis of Algorithms	Major Core	4		Capstone Project – I	Project	3
	Programming in Python	Major Core	2		Open Elective-I	Open Elective	3

	Competitive Coding-II	Skill Enhancement Course	1		Open Elective-II	Open Elective	3
	Professional Elective -II	Minor Elective	1		Environmental Science and Management	Value Education	2
	Professional Elective -III	Minor Elective	3		Industrial Internship	Internship	1
	Technical Internship	Internship	1		Professional Ethics	Ability Enhancement Course	1
	RIE-3	Skill Enhancement Course	1		Competitive Coding-IV	Skill Enhancement Course	1
	RIE-4	Ability Enhancement Course	1		Introduction to SDGs	Value Education	1
	General Proficiency-IV	Skill Enhancement Course	1				
	RIE-3						
	Employability - Soft Skills-III	Skill Enhancement Course	1				
	Innovation - Finance & Economics Fundamentals	Skill Enhancement Course	1				
	Research - Emerging Research Tools and Technologies	Skill Enhancement Course	1				
	RIE-4						

	Employability - Aptitude-III	Ability Enhancement Course	1				
	Innovation - Business Strategies & Operations	Ability Enhancement Course	1				
	Research - Statistical Data Analysis	Ability Enhancement Course	1				
Semester-6	System Design	Major Core	3	Semester-8	AI and NLP	Minor Core	4
	Identity And Access Manager	Major Elective	4		Capstone Project – II	Project	9
	Theory of Computation	Major Core	3		Open Elective-III	Open Elective	3
	Professional Elective-IV	Minor Elective	3				
	Professional Elective-V	Minor Elective	3				
	Mid-Grade Project-2	Project	1				
	RIE-5	Skill Enhancement Course	1				
	RIE-6	Ability Enhancement Course	1				
	Competitive Coding-III	Skill Enhancement Course	1				
	Full Stack Development-II	Major Core	2				
	Software Engineering	Major Core	2				
	RIE-5						
	Employability - Soft Skills-IV	Skill Enhancement Course	1				
	Innovation - Business Planning & Legal Basics	Skill Enhancement Course	1				

	Research - Domain specific Term Paper	Skill Enhancement Course	1				
	RIE-6						
	Employability - Aptitude-IV	Ability Enhancement Course	1				
	Innovation - Customer Engagement &	Ability Enhancement Course	1				
	Organic Growth						
	Research - AI & Data-Driven Research	Ability Enhancement Course	1				

Elective Basket Name			
Semester	Course Name	Course Category	Credits
4	Introduction to Shell Programming	Minor Elective	4
4	Virtualization in Cloud Computing	Minor Elective	4
5	Cloud Computing Architecture	Minor Elective	1
5	Configuration Management and Infrastructure as Code	Minor Elective	1
5	Principles of Virtualization	Minor Elective	3
5	Microservices Architecture & Implementation	Minor Elective	3
5	Computational Data Analytics	Minor Elective	3
6	Web Services	Minor Elective	3
6	Data Warehousing and Data Mining	Minor Elective	3
6	Fog and Edge Computing	Minor Elective	3
6	Cloud Design Pattern	Minor Elective	3
6	Advanced Machine Learning and Mathematical Modelling	Minor Elective	3
7	Web Framework in Cloud (Angular JS)	Minor Elective	3
7	Privacy and Security in IOT	Minor Elective	3

4th Year Curriculum Design Framework [For Hons. / Hons. with Research Tracks of Four Year Undergraduate Programs (FYUP)]

Extending Expertise	Course Category	Credits	Contextualizing Learning	Course Category	Credits	Experiential Learning	Course Category	Credits
Open Elective-I	Open Elective	3	Environmental Science and Management	Value Education	2	Capstone Project – I	Project	3
Open Elective-II	Open Elective	3	Professional Ethics	Ability Enhancement Course	1	Capstone Project – II	Project	9
Open Elective-III	Open Elective	3	Introduction to SDGs	Value Education	1	Industrial Internship	Internship	1
Privacy and Security in IOT	Minor Elective	3				Competitive Coding- IV	Skill Enhancement Course	1
Web Framework in								
Cloud (Angular JS)								
Blockchain	Minor Core	4						
AI and NLP	Minor Core	4						